MATH 152 **QUIZ 3A** Name Solutions

Mrs. Bonny Tighe 25 points

6.5,7.1 TA_____Wed 9/20/23

3pts 1. Find the average value of $f(x) = \dfrac{\cos x}{1+\sin^2 x}$ *on* $[0, \pi/2]$

$$\frac{1}{\pi/2 - 0}\int_0^{\pi/2} \frac{\cos x}{1+\sin^2 x}\,dx \qquad u = \sin x \quad du = \cos x\,dx$$

$$\frac{2}{\pi}\int \frac{1}{1+u^2}\,du = \frac{2}{\pi}\tan^{-1}(\sin x)\Big]_0^{\pi/2} =$$

$$\frac{2}{\pi}\left[\tan^{-1}(\sin \pi/2) - \tan^{-1}(\sin 0)\right] = \frac{2}{\pi}\left[\tan^{-1}(1) - \tan^{-1}(0)\right]$$

$$\frac{2}{\pi}\left[\frac{\pi}{4}\right] = 1/2$$

3pts 2. Find the numbers b such that the average value of $f(x) = 5 - 8x + 3x^2$ on the interval $[0, b]$ is equal to 2.

$$\frac{1}{b-0}\int_0^b (5-8x+3x^2)\,dx = \frac{1}{b}\left[5x - 4x^2 + x^3\right]_0^b =$$

$$\frac{1}{b}\left[5b - 4b^2 + b^3 - 0\right] = 5 - 4b + b^2 = 2$$

$$3 - 4b + b^2 = 2$$

$$(3-b)(1-b) = 0 \qquad b = 1, 3$$

3pts 3. In a certain city the temperature (*in* F^o) t hours after is modeled by the function by $T(t) = 50 + \sin\dfrac{\pi t}{12}$. Find the average temperature during the period from 9 am to noon.

$$\frac{1}{3-0}\int_0^3 \left(50 + \sin\frac{\pi}{12}t\right)dt = \frac{1}{3}\left[50t - \frac{12}{\pi}\cos\left(\frac{\pi}{12}t\right)\right]_0^3$$

$$\frac{1}{3}\left[150 - \frac{12}{\pi}\cos\left(\frac{\pi}{4}\right) - \left(0 - \frac{12}{\pi}\cos 0\right)\right]$$

$$\frac{1}{3}\left[150 - \frac{12}{\pi}\left(\frac{\sqrt{2}}{2}\right) + \frac{12}{\pi}(1)\right] \; F^o$$

4. Evaluate the integrals. a) $\int x(\sec^2 x)\,dx$

4pts

$u = x \qquad dv = \sec^2 x\,dx$

$du = dx \qquad v = \tan x$

$x\tan x - \int \tan x\,dx$

$x\tan x - \ln|\sec x| + C$

4pts b) $\int e^{2x}\cos 3x\,dx$

$u = \cos 3x \qquad dv = e^{2x}dx$

$du = -3\sin 3x\,dx \qquad v = \frac{1}{2}e^{2x}$

$= \frac{1}{2}e^{2x}\cos 3x - \int \frac{1}{2}e^{2x}(-3\sin 3x)\,dx$

$= \frac{1}{2}e^{2x}\cos 3x + \frac{3}{2}\int \sin 3x\, e^{2x}\,dx$

$u = \sin 3x \qquad dv = e^{2x}dy$

$du = 3\cos 3x\,dx \qquad v = \frac{1}{2}e^{2x}$

$= \frac{1}{2}e^{2x}\cos 3x + \frac{3}{2}\left[\frac{1}{2}e^{2x}\sin 3x - \int \frac{1}{2}e^{2x}\,3\cos 3x\,dx\right]$

$= \frac{1}{2}e^{2x}\cos 3x + \frac{3}{4}e^{2x}\sin 3x - \frac{9}{4}\int e^{2x}\cos 3x\,dx$

$-9/4$

$+\frac{9}{4}$

$\frac{13}{4}$

$= \frac{4}{13}\left[\frac{1}{2}e^{2x}\cos 3x + \frac{3}{4}e^{2x}\sin 3x\right] + C$ OR

$\frac{2}{13}e^{2x}\cos 3x + \frac{3}{13}e^{2x}\sin 3x + C$

4pts c) $\int p^2 \cos p \, dp$

$u = p^2$ $\quad dv = \cos p \, dp$

$du = 2p \, dp$ $\quad v = \sin p$

$p^2 \sin p - \int \sin p \, 2p \, dp$

$u = 2p$ $\quad dv = \sin p \, dp$

$du = 2 dp$ $\quad v = -\cos p$

$p^2 \sin p - \left[2p(-\cos p) - \int -\cos p \, 2 dp \right]$

$p^2 \sin p + 2p \cos p - 2 \int \cos p \, dp$

$= p^2 \sin p + 2p \cos p - 2 \sin p + C$

4pts d) $\int \tan^{-1}\left(\frac{1}{x}\right) dx$

$u = \tan^{-1}\left(\frac{1}{x}\right)$ $\quad dv = dx$

$du = \frac{1}{1+\left(\frac{1}{x}\right)^2} \cdot \frac{-1}{x^2} dx$ $\quad v = x$

$du = \frac{-1}{x^2+1} dx$

$= x \tan^{-1}\left(\frac{1}{x}\right) - \int \frac{-x}{x^2+1} dx$ $\quad u = x^2+1$, $du = 2x \, dx$

$= x \tan^{-1}\left(\frac{1}{x}\right) + \frac{1}{2} \ln|x^2+1| + C$

MATH 152
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QUIZ 3

25 points
6.5,7.1

Name Solutions

TA_____ Wed. 9/20/23

3pts 1. Find the average value of $f(x) = \dfrac{\sec^2 x}{1+\tan x}$ *on* $[0, \pi/6]$

$u = 1 + \tan x$
$du = \sec^2 x \, dx$

$$\frac{1}{\pi/6 - 0}\int_0^{\pi/6} \frac{\sec^2 x}{1+\tan x}\,dx$$

$$\frac{6}{\pi}\int \frac{1}{u}\,du = \frac{6}{\pi}\ln|1+\tan x|\Big]_0^{\pi/6} =$$

$$\frac{6}{\pi}\left[\ln\left|1+\sqrt{3}/3\right|\right]$$

3pts 2. The linear density in a rod 2 m long is $\dfrac{10x}{\sqrt{x^2+1}}\ kg/m$ where x is measured in meters from one end of the rod. Find the average density of the rod.

$u = x^2+1$
$du = 2x\,dx$
$\frac{du}{2x} = dx$

$$\frac{1}{2-0}\int_0^2 \frac{10x}{\sqrt{x^2+1}}\,dx = \frac{1}{2}\int_0 \frac{10x}{u^{1/2}}\cdot\frac{du}{2x}$$

$$\frac{10}{4}\int u^{-1/2}\,du = \frac{5}{2}\cdot\frac{1}{1/2}u^{1/2}\Big] = 5\sqrt{x^2+1}\Big]_0^2 =$$

$$5\left[\sqrt{5}-\sqrt{1}\right]$$

3pts 3. Find the numbers b such that the average value of $f(x) = 3x^2-12x+13$ on the interval $[0, b]$ is equal to 5

$$\frac{1}{b-0}\int_0^b (3x^2-12x+13)\,dx = \frac{1}{b}\left[x^3-6x^2+13x\right]_0^b =$$

$$\frac{1}{b}\left[b^3-6b^2+13b-0\right] = b^2-6b+13 = 5$$

$b^2-6b+8=0$
$(b-4)(b-2)=0$

$b = 2, 4$

4pts 4. Evaluate the integrals. a) $\int x(\csc^2 x)dx$

$u = x$ $\quad dv = \csc^2 x\, dx$

$du = dx$ $\quad v = -\cot x$

$-x\cot x - \int -\cot x\, dx$

$-x\cot x + \ln|\sin x| + C$

4pts b) $\int \tan^{-1}\left(\frac{1}{x}\right)dx$

$u = \tan^{-1}(1/x)$ $\quad dv = dx$

$du = \frac{1}{1+\left(\frac{1}{x}\right)^2}\left(-\frac{1}{x^2}\right)dx$ $\quad v = x$

$x\tan^{-1}\left(\frac{1}{x}\right) - \int x\left(\frac{1}{1+\frac{1}{x^2}}\cdot\left(-\frac{1}{x^2}\right)\right)dx$

$x\tan^{-1}\left(\frac{1}{x}\right) - \int \frac{-x}{x^2+1}dx$

$u = x^2+1$

$du = 2x\,dx$

$x\tan^{-1}\left(\frac{1}{x}\right) + \frac{1}{2}\ln|x^2+1| + C$

4pts c) $\int e^{3x}\sin 2x\,dx$

$u = \sin 2x \qquad dv = e^{3x}dx$

$du = 2\cos 2x\,dx \qquad v = \frac{1}{3}e^{3x}$

$= \frac{1}{3}e^{3x}\sin 2x - \int \frac{1}{3}e^{3x}2\cos 2x\,dx$

$= \frac{1}{3}e^{3x}\sin 2x - \frac{2}{3}\int e^{3x}\cos 2x\,dx$

$u = \cos 2x \qquad dv = e^{3x}dx$

$du = -2\sin 2x\,dx \qquad v = \frac{1}{3}e^{3x}$

$= \frac{1}{3}e^{3x}\sin 2x - \frac{2}{3}\left[\frac{1}{3}e^{3x}\cos 2x - \int \frac{1}{3}e^{3x}(-2\sin 2x)\,dx\right]$

$= \frac{1}{3}e^{3x}\sin 2x - \frac{2}{9}e^{3x}\cos 2x - \frac{4}{9}\int e^{3x}\sin 2x\,dx$

1

$+\frac{4}{9}$

$\frac{13}{9}$

$= \frac{9}{13}\left[\frac{1}{3}e^{3x}\sin 2x - \frac{2}{9}e^{3x}\cos 2x\right] + C$

4pts d) $\int t^2 5^{-2t}\,dt$

$u = t^2 \qquad dv = 5^{-2t}dt$

$du = 2t\,dt \qquad v = -\frac{1}{2\ln 5}5^{-2t}$

$-\frac{t^2}{2\ln 5}5^{-2t} - \int -\frac{1}{2\ln 5}5^{-2t}(2t)\,dt$

$-\frac{t^2 5^{-2t}}{2\ln 5} + \frac{1}{\ln 5}\int t\,5^{-2t}dt$

$u = t \qquad dv = 5^{-2t}dt$

$du = dt \qquad v = \frac{5^{-2t}}{-2\ln 5}$

$-\frac{t^2 5^{-2t}}{2\ln 5} + \frac{1}{\ln 5}\left[\frac{t5^{-2t}}{-2\ln 5} - \int -\frac{1}{2\ln 5}5^{-2t}dt\right]$

$-\frac{t^2 5^{-2t}}{2\ln 5} - \frac{t5^{-2t}}{2(\ln 5)^2} + \frac{1}{2(\ln 5)^2}\int 5^{-2t}dt$

$-\frac{t^2 5^{-2t}}{2\ln 5} - \frac{t5^{-2t}}{2(\ln 5)^2} - \frac{1}{4(\ln 5)^3}5^{-2t} + C$